Impact energy and TNT equivalents

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Convert an energy estimate without claiming a complete damage prediction.

SUCCESS CRITERIA - I can...

1. Convert speed units before squaring.
2. Calculate kinetic energy from stated model values.
3. Distinguish an energy equivalent from an explosion or damage forecast.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “equivalent” and “kinetic energy” to explain your model or result. Sentence starters are optional.

equivalent: equal in the specified quantity, not necessarily in other effects

kinetic energy: energy associated with motion

impact: a collision or strike between objects

01 Bring back an earlier idea

Two students perform the same lift. One takes 3 times as long as the other.
Compare work and average power.

Word help: power = rate of energy transfer

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K = \frac{1}{2}mv^2$; 1 ton TNT equivalent = 4.184×10^9 J. Equivalent energy does not mean identical effects.

Worked reasoning: A fictional event transfers 8.4×10^6 J. Using 4.2×10^6 J per kg TNT as an energy comparison gives 2 kg TNT equivalent. This compares energy, not identical effects.

Your turn: Draw a labeled energy comparison and name one effect this number alone cannot predict.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

03 Look and explain

A report gives impact energy in J and another in TNT-equivalent tons. Are they necessarily describing different physical quantities?

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Build the solution

A supplied energy is 8.368×10^9 J. Divide by 4.184×10^9 J/ton.
Find TNT equivalent.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Why should an energy-equivalent number not be reported as a precise area of destruction?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

Lift a 5 kg object by 7 m.
Use g=10 N/kg.
Find the change in gravitational potential energy.

Word help: potential energy = energy associated with the configuration of interacting objects

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

At fixed mass, speed halves. What happens to K and its TNT equivalent?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two events have equal energy but spread it over different areas.
Explain why local effects could differ without changing total energy.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3